Open-Ended Plural Intelligence:
Individuation, Conserved Distinctions, and Relational Coordination

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Abstract

An open-ended intelligence must act through bounded observations without mistaking those observations for the world, retain its organization while changing, and coordinate with agents whose views need not converge. We present a reusable formal architecture for these requirements. Its primitive objects are typed completion fibres, observer-indexed distinctions, proof-relevant relations between views, and recoverable evidence. Cardinal variety, functional translations, compact contexts, and executable fast paths are derived only when the relevant witnesses exist.

The central no-go result says that a fixed context which remains sufficient for an accumulated state-separating query schedule must be injective. Consequently, no genuinely lossy permanent summary is sufficient for every future policy of an open-ended agent. The constructive counterpart is a canonical weakest sufficient view for every finite active query family, refreshed from recoverable evidence rather than recursively summarized. We connect this result to formal accounts of distinction conservation, relational closure, processual individuation, variety-under-constraint, metasystem transitions, stigmergy, protected freedom, and source-scoped adaptive realization.

The development is mechanized in Lean 4. Positive witnesses and destructive controls distinguish exact presentations from lossy readouts, relational transport from earned functionality, individuation from static closure, and semantic admission from profitability or cache residence. The resulting theory is general mathematics; Prime DTT, GSLT-IL, MeTTa spaces, WM-style world models, and self-modifying cognitive architectures enter as theorem-backed instances rather than defining the theory.

1 The problem: bounded views in an unbounded life

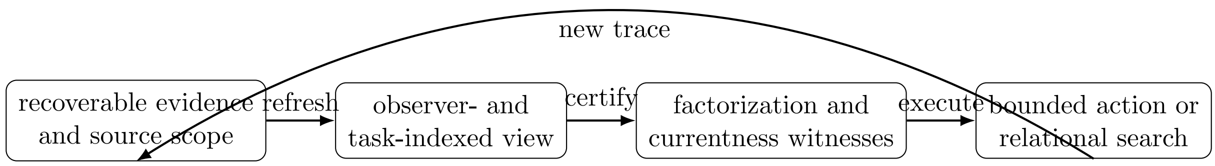
A cognitive system never carries the whole world in its active context. It sees a view, acts, receives new evidence, changes its policies, and often changes the very language in which it describes the problem. Three tempting simplifications are therefore unsafe:

(1) treating one compact summary as permanently sufficient;

(2) identifying a relation between viewpoints with a function before totality and determinism have been established;

(3) identifying the persistence of an agent with the persistence of one process, prompt, model, or data representation.

The alternative developed here starts from informative objects and derives their lossy readouts. The main dependency direction is



This is not a proposal to store all history in every hot path. It separates an authoritative, persistent source from the small view needed by the present task. When a policy changes, the system either proves that the policy still factors through the current view, or refreshes from the source, or uses an exact fallback.

The intellectual sources are complementary. Heylighen's distinction conservation, relational closure, constrained variety, metasystem transitions, and stigmergy supply formal questions about loss, level formation, and mediated coordination [1, 2, 3, 4]. Weinbaum's account of becoming and Weinbaum–Veitas's account of open-ended intelligence motivate treating individuation as a process rather than a static label [5, 6, 7]. Veitas and Weinbaum's "world of views" motivates plural coordination without an assumed global representation [8, 9]. The definitions and bridge theorems below are our formal constructions; citations identify the source of the motivating idea, not a claim that the source already proved the Lean theorem.

2 Informative objects before numerical readouts

2.1 Typed completion fibres

Let W be a type of worlds and let a statement be a predicate $s : W \to \mathsf{Prop}$. Its compatible completions are retained as a type:

Definition 2.1 (Completion fibre).

$$\mathsf{Completion}(s) \;:=\; \{w : W \mid s(w)\}.$$

This type contains the compatible worlds themselves. A finite cardinality such as $|\mathsf{Completion}(s)|$ is a derived observation. If s_1 is stronger than s_2, every completion of s_1 embeds into the completions of s_2. The Lean development proves this contravariant embedding before proving its finite cardinal consequence.

The order matters. Two statements can have equally many completions while admitting different completions. A number cannot transport a future witness, explain why an option is available, or retain provenance. The fibre can.

Proposition 2.2 (Exact presentation). *If two presentations of a completion problem are related by an equivalence that preserves the underlying membership judgment, then their completion fibres are equivalent. When the fibres are finite, their cardinalities are equal.*

This is the formal basis for comparing presentations without asserting that every quantity is representation-independent.

2.2 Observer-indexed variety

An observer is a function $o : W \to V$. It distinguishes two worlds when their observations differ, and its informative variety is the image type

$$\mathsf{Variety}(o) := \{v : V \mid \exists w,\ o(w) = v\}.$$

The fibre above v is $\{w : W \mid o(w) = v\}$. Thus the theory retains both what the observer sees and which states the observation identifies.

Theorem 2.3 (Observer relativity). *There is no observer-independent finite cardinality of variety in general. In the formal control, a micro-observer sees both Boolean coordinates and a macro-observer forgets the second. Their observed cardinalities are four and two, respectively.*

This theorem is intentionally small. Its role is to prevent a large architectural error: a later "variety" score must name the observer and domain over which it was measured.

3 Distinction conservation as a semantic gate

For source and target distinction relations D_S and D_T, a map $f : S \to T$ conserves distinctions when

$$D_S(x, y) \Longrightarrow D_T(f(x), f(y)).$$

When both distinctions are inequality, distinction conservation is exactly injectivity. This recovers a precise core of Heylighen's causality-as-distinction- conservation proposal without identifying every causal question with injectivity [1].

The useful case is observer-indexed. A transform may safely erase differences that a declared observation cannot see, but it may not erase differences on an observable that a later policy uses. This makes the gate task-relative without making it arbitrary.

3.1 Answer effects

The GSLT execution layer already distinguishes an answer representation from its effect on an answer container. The formalization proves:

Theorem 3.1 (Faithfulness criterion)**.** *For an answer effect, conservation of answer distinctions is equivalent to the existing independently defined faithfulness property.*

The equivalence prevents a new slogan from becoming a parallel foundation. It also classifies two real losses:

- list-to-bag conversion preserves multiplicity but destroys order;
- bag-to-support conversion preserves truth support but destroys multiplicity.

Concrete collision witnesses prove both failures. By contrast, decorating every answer occurrence with a revision identifier conserves the original distinctions.

Remark 3.2 (No universal "dedup is harmless")**.** *Deduplication may be correct for a support-valued observer and incorrect for a proof-occurrence-valued observer. The admissibility theorem must name the observation; answer-set equality alone is not a universal performance license.*

4 Closure and individuation

4.1 Law-bearing relational closure

A relational-closure system consists of a network domain, a closure operation, and proofs of extensivity, monotonicity, and idempotence. It induces a standard mathematical closure operator. The formal object also retains proof-relevant receipts for connections generated by closure.

This follows the purpose of Heylighen's relational closure: suppress low-level internal distinctions while making higher-level external distinctions available [2]. The Lean theory makes the loss explicit. A seed network and its closure can be distinct presentations that induce the same closed network.

Example 4.1 (Symmetric closure)**.** *Starting from the one-way Boolean edge* $\mathsf{false} \to \mathsf{true}$*, symmetric closure adds the reverse edge. The reverse edge has a generated-connection receipt; it is absent from the seed. The seed and the closed presentation nevertheless have the same closure.*

Thus closure is a lawful abstraction, not an identity claim.

4.2 Individuation is a process

The general individuation theory separates four objects:

1. a theory of system states, boundaries, and admissible steps;
2. a static individuated state satisfying closure at a boundary;
3. a proof-relevant process connecting states through steps;
4. persistence, which relates a process to a protected observation.

Processes compose. "Becoming individuated" requires both a process and an individuated endpoint. Static closure alone does not manufacture a process.

Theorem 4.2 (WM-style instance). *The existing Markov-logic individuation structure, augmented by its explicit non-emptiness premise, is equivalent to an instance of the general static individuation theory. Existing dynamic-individuation paths induce general individuation processes and persistence results under their stated model assumptions.*

This recovers rather than replaces the earlier world-model machinery. It also gives a precise interpretation of Weinbaum and Veitas's proposal: the identity of an open agent is carried by an organized transformation process, not by a frozen state [7, 6].

4.3 When does a process warrant a new generation?

An enactive generation is not inferred from closure alone. It requires a task interpretation that maps the individuation process to a parent/child policy relation and proves the relevant completion inclusion. The development contains both a positive process-to-generation construction and a negative static shell which is closed but supplies no generation witness.

This boundary is important: a hierarchy of abstraction levels should be earned by a semantic transition, not introduced merely because a new type name is convenient.

5 Constraint, variation, and metasystem transitions

A constraint on states is a predicate; its fibre is the subtype of states satisfying it. Refinement of constraints induces an embedding of fibres. Separately, a proof-relevant relation between configurations is classified by totality and endpoint determinism. This yields independent axes for:

$$\text{constraint present/absent} \times \text{static variety/unitarity} \times \text{dynamic variation/rigidity.}$$

The eight cases are represented by constructors rather than inferred from prose. The theory supplies a positive organization with constraint, static variety, and dynamic variation, and a null organization lacking all three. This mechanizes the classification proposed in Heylighen's treatment of systems as constraints on variety [3].

Definition 5.1 (Metasystem transition). *A metasystem packages lower-level systems, admissible variation among them, a higher-level constraint on that variation, and a task interpretation explaining why the resulting organization constitutes a new generation.*

The completion fibre of the higher constraint is equivalent to the admissible higher-level configurations. A concrete transition is constructed. A separate negative theorem shows that mere aggregation plus an external selector is insufficient: it provides neither internally organized variation nor the task witness needed for level introduction.

This is a conservative use of the historical idea. We do not claim that every evolutionary metasystem transition is captured by the present schema; the schema isolates a theorem-level condition usable by language and agent architectures.

6 A world of views: relation first, function earned

Suppose each viewpoint i has a state type S_i. A coordination route from i to j is taken primarily to be proof-relevant:

$$R_{ij} : S_i \to S_j \to \mathsf{Type}.$$

It may be partial, branching, or carry multiple witnesses between the same endpoints. A functional translation $S_i \to S_j$ is derived only when the relation is total and proof-relevantly endpoint-deterministic.

Theorem 6.1 (Representability boundary)**.** *A functional transport exists exactly when every source has a target and all proof-relevant derivations from a source agree on the target. The selected function retains a witness that its graph lies in the original relation.*

The plural example has one request view and two distinct compatible responses. It is open and coordinating, yet admits no functional request-to-response map. Its reverse route is functional. This makes non-convergence a positive architecture rather than a failed attempt at translation, following the direction proposed by Veitas and Weinbaum [8, 9].

6.1 Local coherence need not glue globally

The cognitive-architecture bridge uses real rational pair charts. It constructs pairwise-compatible local marginals which have no global classical joint realization.

Theorem 6.2 (No total gluer from local coherence)**.** *There is no function which turns every locally coherent family in the example into a global realizing chart.*

This does not license inconsistency by fiat. It says that the right output of a failed gluing problem is a contextual family with its local witnesses and obstruction, not fabricated global consensus.

7 Stigmergy as trace-mediated coordination

Following Heylighen, the formal interface names five components: agent, action, medium, trace, and coordination [4]. A delayed stigmergic episode requires an earlier action to leave a persistent trace in the medium and a later action to depend on that trace.

The theory deliberately includes Bateson's natural control: a medium may permit direct coordination while failing to expose any persisted trace. Such a medium cannot construct the stipulated delayed stigmergic episode. Therefore the definition does not collapse all coordination into stigmergy.

The MeTTa instance uses an actual proof-relevant space. Publishing an occurrence leaves a trace; a later query returns a derivation witnessing that occurrence. The space is consequently a coordination medium, not merely a set-valued store. Order, multiplicity, and provenance remain observable where the query discipline requests them.

8 The open-ended context theorem

Let S be the recoverable evidence state, Q a type of queries, and A a type of answers. A query q denotes $q : S \to A$. For a finite active set $F \subseteq Q$, define the canonical active view

$$\pi_F(s) := (q(s))_{q\in F} \in \prod_{q\in F} A.$$

Theorem 8.1 (Weakest sufficient finite view)**.** *The active-answer vector π_F is sufficient for every query in F. If another view $v : S \to V$ is sufficient for every query in F, then π_F factors through v. Thus π_F is the coarsest, or weakest, view sufficient for the declared query family.*

When A and F are finite, the representation has the exact ambient bound

$$|A|^{|F|}.$$

This is not a claim that all answer vectors occur; it is the size of the canonical function space.

Now let $F_0 \subseteq F_1 \subseteq \cdots$ be an accumulated query schedule. Say it separates states when every unequal pair is distinguished at some epoch.

Theorem 8.2 (No permanent lossy context)**.** *If a fixed view $v : S \to V$ remains sufficient at every epoch for an accumulated state-separating query schedule, then v is injective. Hence no non-injective fixed context is permanently sufficient for an open-ended separating policy family.*

The result permits a different decoder at every epoch. The obstruction is not a poor decoder; it is permanent information loss.

Example 8.3 (Unbounded evidence, two-valued epoch views)**.** *Take $S = \mathbb{N}$ and one Boolean bit-query per epoch. Every individual epoch view has exactly two possible answers, while the accumulated schedule separates every natural number. A constant permanent context is formally impossible as a sufficient view, even though each current query is tiny.*

The constructive architecture retains S as recoverable evidence and regenerates π_F whenever F changes. It never recursively summarizes a summary. This explains how bounded active context and open-ended intelligence can coexist.

9 Three freedoms and protected self-transformation

The formalization keeps three notions apart:

$$\begin{aligned}
\text{semantic freedom} &:= \mathsf{Completion}(p),\\
\text{epistemic latitude} &:= \{s' \mid o(s') = o(s)\},\\
\text{executable freedom} &:= \{c : \mathsf{Completion}(p) \mid c \text{ has a current receipt}\}.
\end{aligned}$$

More compatible refinements can be valuable; more states hidden by an observer can be mere ignorance; more currently executable actions can be unsafe. None of the three numbers subsumes the other two.

A self-modification carries a protected family P of completions. A valid transformation supplies a map from every protected pre-modification completion to a post-modification completion, plus current authority. Proof-backed weakening yields such a map constructively. The Gödel-style architecture bridge proves protected observation agreement using its existing dynamic-individuation assumptions.

Proposition 9.1 (Unconstrained cardinality is unsafe)**.** *Maximizing the number of completions without a protected family can prefer a vacuous or unsafe weakening. A concrete finite counterexample has strictly more completions after weakening while losing a declared protected property.*

Thus option preservation is relative to a correctness fibre and a protected family, not a universal instruction to maximize nondeterminism.

10 Source-scoped adaptive native realization

Fast execution is licensed by four separate judgments:

1. *semantic sufficiency*: the requested policy factors through the key;

2. *distinction contract*: the key conserves the distinctions named by the requested observation;

3. *currentness*: the source and authority revisions still match;

4. *profitability and retention*: preparation is worth its cost and the plan is resident.

Only the first three affect semantic activation. Profitability may rank or select an already valid realization; it cannot mint truth. Cache residence is an operational fact, not a semantic verdict.

Definition 10.1 (Semantic admission)**.** *A prepared realization is semantically admissible when its key supports the policy, its explicit source/key relation conserves the requested distinctions, its observable result agrees with the raw result, and an exact raw fallback remains available.*

The Prime/GSLT bridge proves that a current source-scoped receipt induces semantic admission. The Cost bridge proves that the identity key on a proof-relevant elaboration supports every request. Negative controls show:

- a constant key can support a constant policy while failing exact distinction conservation;
- support erasure identifies a one-occurrence bag with a two-occurrence bag;
- a stale revision blocks activation but leaves exact fallback available;
- overlapping evidence sources do not by themselves establish independence.

This yields the implementation shape

immutable admitted plan: typed program, source scope, proof-DAG root, authority revision, prepared frame layout

$\downarrow$

private invocation: input, dense slots, undo-trail mark, demanded views, effect/provenance root

$\downarrow$

stale, unsupported, or unprofitable: relational search or exact raw fallback

The mathematics does not prescribe a cache policy. It states what any cache policy must preserve.

11 Open-system extension

An open cognitive language must admit new declarations, rules, views, and execution realizations without silently changing old meanings. The GSLT extension boundary therefore separates authored composition from validated composition.

Theorem 11.1 (Current validated growth)**.** *A validated presentation extension with a current dependency receipt transports every old checked judgment while preserving its protected goal and exact derivation.*

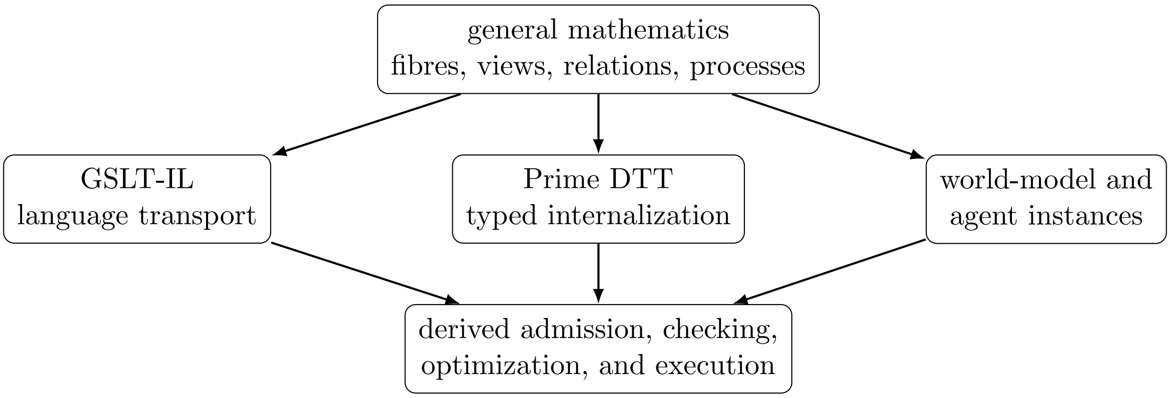
Two layers may be individually admitted and their authored merge may be syntactically available while the combined system is invalid. A validated composite consequently requires revalidation of the merge.

Theorem 11.2 (Boundary crossing requires revalidation)**.** *There exist individually admitted layers whose authored merge is incompatible and has no validated-composite witness. Failure to validate the extension does not refute the source system.*

This is the conservative-growth rule for Prime/GSLT: old checked meaning is transported by proof; new joint structure earns admission at the boundary. The construction has no finite-vocabulary premise.

12 Consequences for Prime, GSLT-IL, and cognitive architectures

The general theory imposes the following dependency order.



Prime is not defined as an authority checker. It is a candidate internal language for the typed completion, relation, evidence, and protected-transformation structures. GSLT-IL provides proof-relevant language routes, with functions as represented special cases. NIK-style authority, compilation, caching, and cost are downstream consumers. CeTTa or another runtime may implement the settled semantics directly and establish conformance without becoming the definition of the theory.

For cognitive architectures, identity is the continuing organization of protected commitments, authoritative memories and receipts, executable capabilities, and the process that reconstructs task-relative views. A restart may preserve that organization; a continuously running process may lose it. Multi-agent spaces retain local views and compatibility witnesses, glue only when a certificate exists, and use persistent traces for delayed coordination.

13 What has and has not been established

The mechanized development establishes the theorem families summarized in Table 1. It does not establish a universal theory of intelligence, a globally optimal view-selection policy, or a complete operational adequacy theorem for every runtime.

Open obligation 13.1 (Operational adequacy)**.** *Relate the abstract source-scoped realization receipts to complete runtime receipts, including work, span, allocation, invalidation, and proof-DAG sharing, without erasing answer occurrences.*

Open obligation 13.2 (Contextual invention)**.** *Extend weakest-sufficient views to open terms and policies under binders. This needs contextual codes and substitution coherence; the closed finite-query theorem does not by itself solve typed predicate invention.*

Open obligation 13.3 (Certified parallel composition)**.** *Prove when completion fibres of parallel agents compose as products. Disjoint source scope alone is insufficient; resource and communication independence must also be certified. Interacting systems generally require a pullback or proof-relevant relational convolution.*

Theme	Positive result	Destructive control
Completion and variety	typed fibres; exact-presentation equivalence	equal counts need not identify options; observer cardinalities differ
Distinction conservation	equivalence with answer-effect faithfulness	order and multiplicity erasures collide
Closure and individuation	law-bearing closure; composable becoming process	static closure does not create a generation
Metasystems	constrained variation plus task interpretation	aggregation and external selection are insufficient
World of views	relational coordination; represented functional route	branching route has no function; local coherence need not glue
Stigmergy	proof-relevant space publication/query episode	ephemeral direct coordination lacks delayed trace mediation
Open-ended context	weakest active view; permanent sufficiency implies injectivity	constant context fails the separating schedule
Protected freedom	protected completion maps under valid modification	cardinal freedom alone can discard a protected property
Adaptive realization	sufficiency, distinction, currentness, fallback	constant key, stale receipt, support erasure, source overlap
Open extension	exact old-judgment transport under validated growth	individually valid layers can have an invalid merge

Table 1: Theorem-backed positive and negative boundaries.

Open obligation 13.4 (Selection under distribution shift)**.** *Compare weakness, description length, and probabilistic selectors while retaining candidate occurrences, distinct support, runtime cost, and future-task success as separate measurements.*

14 Conclusion

Open-endedness does not require an infinitely large active context or a single global world model. It requires that bounded views remain honest about what they forget, that richer evidence can be revisited, that translations remain relational until functionality is earned, and that self-transformation preserves explicitly protected futures.

The core lesson is simple. Keep the informative object first: the completion fibre, the proof-relevant route, the process, the trace, the receipt. Derive the number, Boolean, function, cache key, or compact view only with a theorem describing the loss. That discipline supports plural open systems while still permitting efficient native execution.

A Lean theorem map

The implementation is organized by semantic subject rather than by application. The principal modules are:

- `Mettapedia/Enactive/CompletionFibre.lean`, `Mettapedia/Cybernetics/ObservedVariety.lean`;
- `Mettapedia/Cybernetics/DistinctionConservation.lean`, `Mettapedia/GSLT/Dynamics/AnswerDistinctionConservation.lean`;
- `Mettapedia/Cybernetics/RelationalClosure.lean`, `Mettapedia/Cybernetics/Individuation.lean`, `Mettapedia/Logic/MarkovLogicIndividuationBridge.lean`, `Mettapedia/Enactive/IndividuationGeneration.lean`;
- `Mettapedia/Cybernetics/ConstrainedVariety.lean`, `Mettapedia/Enactive/MetasystemTransition.lean`;

- `Mettapedia/GSLT/Dynamics/WorldOfViews.lean`, `Mettapedia/CognitiveArchitecture/Agent/WorldOfViewsBridge.lean`;
- `Mettapedia/Cybernetics/Stigmergy.lean`, `Mettapedia/Languages/MeTTa/StigmergicSpace.lean`;
- `Mettapedia/CognitiveArchitecture/Agent/OpenEndedContext.lean`;
- `Mettapedia/Enactive/ProtectedFreedom.lean`, `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/ProtectedFreedomBridge.lean`;
- `Mettapedia/Languages/MeTTa/Prime/SourceScopedAdaptiveRealization.lean`;
- `Mettapedia/GSLT/LanguageDef/OpenSystemExtensionBoundary.lean`.

The historical claims in the paper are not inserted as axioms. They motivate definitions; all named project results are checked Lean declarations with explicit premises.

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